

Splitting the Mind of Zeus

How crucial aspects of logic arise, like Pallas Athena,
from splitting the focus of attention from its context

*Realizability, collection monads, and the choice content of the McBride derivative in a
finite-limits metalanguage*

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Abstract

We set out to make visible the seams along which a logic is assembled. The commitment that makes the seams visible is a realizability semantics: a formula denotes the collection of computations that witness its computational content. Under that commitment a logic factors into three components — a *collection* component (how witnesses are aggregated), a *structural* component (term constructors lifted to predicates on witnesses), and a *behavioral* component (contexts labeling transitions lifted to modalities). The collection component sits awkwardly with categorical logic, because the subobject classifier bakes in *set*-shaped collection: there is no list of all lists. We reconcile the two by going syntactic. Treating every collection as a term type, we use the McBride derivative to form the object of all splittings $(\partial T, T)$ — the *proto power object* — and then ask what must be added to the ambient metalanguage to *package* the splittings back into the collection type. Committing the metalanguage to finite limits (cartesian logic) is what keeps the whole apparatus syntactic: there is no subobject classifier to lean on, so the derivative route is the only route; and choice cannot be asserted as a proposition, so it can enter only as a named section operation. The central claim is that packaging is exactly a section of the quotient presenting the collection monad, that its choice content equals the assertion that the atom object admits a linear order, and that going syntactic discharges this because term positions are already ordered. The construction is kept parametric in the order on atoms, isolating the choice residue at the atom boundary — which is where later reflective (knot-tying) constructions will act. Throughout we note that the equations one quotients by — commutativity and idempotence — are exactly the equations the no-go theorems for distributive laws turn on, so the distributive-law obstruction and the packaging cost are one datum read twice.

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1 Why realizability, and why this matters for the argument to the AI community

Most accounts take a logic as given: a fixed vocabulary of connectives, a notion of model, and a truth predicate that floats above computation. Truth values arrive from a Boolean or Heyting algebra that is postulated, not built. For a discipline that wants its reasoning systems to be inspectable — traceable to constructions whose behavior can be run, audited, and held accountable — this is exactly the wrong starting point. The glue that binds truth to the algebra is opaque, and once it is in place one can no longer see what the logic is *made of*.

These stakes are not specific to artificial intelligence, but they are highly relevant to it. AI advances a value proposition: that important and useful aspects of (human) intelligence can be faithfully rendered as computation. Logic has long been cast as the calculus of reason —

the very paradigm of the rational faculty. To exhibit logic itself as *arising from* computation is therefore direct support for that proposition, since the calculus of reason is thereby shown to be, at bottom, computational. But the philosophical vindication is the smaller part. The larger part is that a computational account of logic yields *effective procedures*: procedures for *generating* logics and for *generalizing* logic-based algorithms. If the connectives are operations of three named kinds (§2), then a new logic is obtained by turning the dials, and an algorithm defined over one logic can be transported, mechanically, to another by re-instantiating the same three components. The decomposition is not a description of logic arrived at after the fact; it is a construction kit. That generative, effective character — not the slogan that reason is computation — is what this research programme offers in support of AI’s value proposition.

So we invert the usual starting point. We want a logic whose every connective is an operation on computations, so that the entire edifice is a construction and nothing is postulated except the constructions themselves. The commitment that delivers this is realizability. Rather than asking whether a formula is true, we ask which computations *witness* it, and we let the formula denote the collection of its witnesses:

$$\llbracket \varphi \rrbracket = \{ e : e \Vdash \varphi \}.$$

This is the Kleene move [1, 2], but with the aggregating container promoted from an incidental detail to a *parameter* of the construction. Once witnesses are computations and formulae are collections of them, every connective is forced to be an operation of one of exactly three kinds, corresponding to the three things one can vary: how witnesses are collected, how they are built, and how they move. This is the decomposition we develop in §2.

There is a second, sharper commitment, and it is the one we most want to defend to the AI community. We insist that the ambient metalanguage — the language in which the collections, the constructions, and the packaging all live — be a *finite-limits* (cartesian, essentially-algebraic) theory [12, 13, 14]. The reason is not minimalism for its own sake. It is that a finite-limits metalanguage is the maximal commitment to “everything is a construction”:

- there are no truth values floating in from a Boolean algebra, because there is in general no subobject classifier (a lex category need not be a topos);
- there is no non-constructive existential, because the only existentials one may assert are provably *unique* ones, carved out as partial operations on equationally-defined subdomains;
- and — the point that will do the real work in §6 and §8 — choice cannot be *stated* as a proposition at all. There is no $\forall$, and $\exists$ is available only when the witness is unique. So the only way to “add choice” is to add a *function symbol* to the signature, together with equations making it a section. In a finite-limits metalanguage, choice is not a truth one assumes; it is an operation one names.

This is exactly the inspectability we are after. When choice enters, it enters as a symbol you can point at, with a documented arity and documented equations, and you can localize precisely where in the construction it was spent. Nothing is hidden in a semantic ether. The seams are visible because we have refused the glue — the subobject classifier — that ordinarily hides them.

The remainder of the note carries this out. §2 gives the three-component decomposition. §3 shows that the collection component is a substructural dial whose settings are governed by two equations, commutativity and idempotence. §4 states the disconnect with categorical logic, and §5 shows that the same two equations are what the no-go theorems for distributive laws

turn on — so the obstruction to a uniform collection component and the cost of packaging are, as we will see, one datum read twice. §6 cashes out the finite-limits commitment. §7 builds the proto power object from the McBride derivative, and §8 defines packaging as a section and states the choice conjecture, parametric in the order on atoms. §9 places the conjecture in the Weihrauch lattice, §10 works the three instances, and §12 draws out the consequences, including the absence of any classicality collapse and the reflective use of the atom-order parameter.

2 Three components of a logic under realizability

Fix a representation of computation. Throughout we take it to be a term rewrite system: terms are computational states, and rewrites are steps. Write T for the type of terms. Fix also a way of aggregating witnesses: a *collection type* S , which will always carry the structure of a monad (§3). A formula denotes an object $\llbracket \varphi \rrbracket : S(T)$ — a collection of terms.

Under these two fixings, three families of connectives arise, and every connective is of one of the three kinds.

Collection connectives. Any operation on S lifts to a connective. If witnesses are collected in sets, the semilattice operations of $S = \mathbf{P}_{\text{fin}}$ give the familiar connectives:

$$\llbracket \varphi \wedge \psi \rrbracket = \llbracket \varphi \rrbracket \cap \llbracket \psi \rrbracket, \quad \llbracket \varphi \vee \psi \rrbracket = \llbracket \varphi \rrbracket \cup \llbracket \psi \rrbracket.$$

This is where the choice of container is felt: change S from sets to multisets, lists, trees, or distributions, and these connectives change with it. The collection component is the parameter we most want to vary, and §3 explains what varying it does.

Structural connectives. Each term constructor of T lifts to a predicate transformer. If $f : T^n \rightarrow T$ is an n -ary constructor, define

$$\llbracket \widehat{f}(\varphi_1, \dots, \varphi_n) \rrbracket = \{ f(e_1, \dots, e_n) : e_i \Vdash \varphi_i \}.$$

The lifted constructor $\widehat{f}$ is a *structural* predicate: it speaks about the *shape* of a witness, asserting that the witnessing computation was built by f from sub-witnesses of the φ_i . In the reflective setting these are the constructors of the rho calculus lifted to namespace predicates [3, 4]; in general they are the introduction half of an Operational-Semantics-in-Logical-Form (OSLF) presentation.

Behavioral connectives. Each rewrite is labeled by the context in which it fires. A step $C[l] \rightarrow C[r]$ carries the one-hole context C , an element of the *derivative* ∂T (§7). Lifting the labels to modalities gives

$$\llbracket \langle C \rangle \varphi \rrbracket = \{ e : e \xrightarrow{C} e', e' \Vdash \varphi \}, \quad \llbracket [C] \varphi \rrbracket = \{ e : \forall e'. e \xrightarrow{C} e' \Rightarrow e' \Vdash \varphi \},$$

the context-indexed Hennessy–Milner modalities. This is the elimination half of OSLF.

Remark 2.1. The structural and behavioral components are the two halves of OSLF, generated respectively from constructors and from labeled transitions. They are already well-understood in the reflective setting. The component that resists the categorical-logic packaging, and that this note is about, is the first one — collection — together with the derivative machinery (∂T) that turns out to glue it to the third.

3 The collection component is a substructural dial

The collection type S is a monad, and the algebraic theory presenting S is not incidental: *its equations are the structural rules of the propositional logic the collection component induces*. This is the first fact that makes the decomposition worth having, because it says the collection component is not “logic plus a data-structure flavor”; it *is* the substructural hierarchy, indexed by how far S sits from set-shaped.

Write the binary aggregation of S as $*$ with unit ε . The correspondence is:

commutativity of $*$	$\longleftrightarrow$	exchange,
idempotence of $*$	$\longleftrightarrow$	contraction,
unit/absorption laws	$\longleftrightarrow$	weakening.

Reading the standard collection monads through this correspondence:

- P_{fin} , the finite-powerset monad — a commutative, idempotent monoid (a join-semilattice) — gives exchange and contraction (and, with the unit, weakening): the full structural regime, hence Heyting/Boolean propositional logic.
- Bag , the finite-multiset monad — the free commutative monoid, commutative but not idempotent — gives exchange without contraction: the multiplicative-linear regime with $\otimes$.
- List , the list monad — the free monoid, neither commutative nor idempotent — gives neither exchange nor contraction: a non-commutative, Lambek-style regime.
- D , the distribution monad — commutative with a convex (barycentric) structure in place of idempotence — gives the probabilistic regime.

Principle 3.1 (Collection is substructure). *The setting of the collection dial is the pair of Booleans (commutative?, idempotent?) of S , and this pair is exactly the presence or absence of exchange and contraction in the induced logic.*

Keep the two coordinates of Principle 3.1 in view. They are the same two coordinates on which the no-go theorems of §5 turn, and the same two equations one quotients by when packaging in §8. Everything in this note is, at bottom, bookkeeping on commutativity and idempotence.

4 The disconnect with categorical logic

Categorical logic packages the collection component through the subobject classifier Ω : the power object of A is Ω^A , the object of subobjects of A . This is a beautiful and total solution — for one setting of the dial. It hard-wires $S = \text{powerset}$. Two things go wrong when we try to honor Principle 3.1 and turn the dial.

First, the classifier fixes the collection shape. There is no Ω -parametrized-by- S ; the power object is subsets, and only subsets. One cannot, inside the Ω story, ask for the multiset or list of witnesses.

Second, and more tellingly, even the operation of forming “the collection of all subcollections as an object of the same shape” fails away from sets. The slogan is: *there is no list of all lists*. To assemble all sublists of a given list into a single *list*, one must linearly order the sublists; there

is no canonical order on them, and choosing one is choice. For sets the analogous assembly is free of order — the subsets form a set with nothing to arrange — which is exactly why the Ω story works for the powerset setting and only there.

Remark 4.1. The disconnect is not a defect of categorical logic to be repaired by a cleverer classifier. It is a signal that the packaging of the collection component into an object of the same shape is *order-laden*, and that the Ω story succeeds precisely by living at the one setting of the dial (powerset) where the order requirement is vacuous. The reconciliation in §7–§8 takes the order requirement seriously and localizes it, rather than wishing it away.

5 The same two equations: distributive laws and their no-go theorems

Before the reconciliation, we want to record why the collection dial cannot be turned freely if one insists on the standard monadic route to the structural connectives, because it is the same two equations again.

The structural connectives of §2, viewed monadically, ask the collection monad S to commute past the term monad T . Lifting a constructor $f : T^n \rightarrow T$ to collections requires a coherent map $(SA)^n \rightarrow S(A^n) \rightarrow SA$, and for this to be coherent across *all* the constructors of T simultaneously one wants a distributive law

$$d : S \circ T \Rightarrow T \circ S,$$

in the sense of Beck [5]. A distributive law is exactly what lets S -collections be pushed through T -structure functorially, yielding the lifted constructors.

The trouble is that such laws are scarce, and the scarcity is governed by commutativity and idempotence. The covariant powerset monad does not distribute over itself [6]; the distribution monad does not distribute over powerset [7]; and the general no-go theorems of Zwart and Marsden [8] show that once the two theories being combined carry equational shapes of the wrong kind — an idempotence or an absorption clashing with the other theory — *no* distributive law can exist. The upshot is that, along the distributive-law route, the only collection monads that commute nicely past term-like T are the set/multiset/distribution-shaped ones: exactly the ones near the top-and-commutative corner of the dial.

Principle 5.1 (One datum, read twice). *The equations that obstruct the distributive law $d : ST \Rightarrow TS$ — commutativity and idempotence — are the same equations that, in §8, one must quotient `List` by to present S , and hence the same equations whose section must be postulated to package. The no-go obstruction and the packaging cost are one datum read twice: once as a law that fails, once as an operation that must be added.*

The reconciliation of §7–§8 is, in effect, a third way around the no-go theorems. The two known escapes are to *quotient* (convex powerset, indexed valuations — adjust S until the equations agree) or to *restrict* (finite, ordered — shrink until the clash cannot arise). The route here is to *add a global structuring operation* — the section — and pay for it in a named, localized axiom. §12 argues these are dual dodges.

6 The finite-limits metalanguage does exactly what we need

Now the metalanguage commitment earns its keep. Take the ambient metalanguage to be a finite-limits theory: the cartesian fragment, with $\top$, $\wedge$, $=$, and existentials only where the witness is provably unique [12, 13, 14]. Absent are $\vee$, $\perp$, $\Rightarrow$, $\neg$, unrestricted $\exists$ and $\forall$, and — in general — a subobject classifier. Three consequences, each of which turns a would-be obstruction into a feature.

(1) There is no Ω to lean on, so the syntactic route is the only route. In a topos one might hope to repair §4 with a cleverer use of the classifier. In a finite-limits metalanguage that hope is simply unavailable: a lex category need not be a topos, and the power object is not on the table. The McBride-derivative construction of §7 is therefore not *an* alternative to the Ω -story; it is the *only* story. The commitment vindicates going syntactic instead of competing with it.

(2) Choice cannot be a proposition, only an operation. In cartesian logic one cannot write the choice principle in $\forall\exists$ form: there is no $\forall$, and $\exists$ may be asserted only under uniqueness. So “adding choice” cannot mean asserting a sentence. It can only mean enlarging the signature with a function symbol and equations that make it a section of some quotient. This is precisely the shape that packaging will take in §8: **pack** is a new operation, and the equation $q \circ \text{pack} = \text{id}$ is choice-as-structure. The metalanguage forces the conjecture into its most honest and most inspectable form.

(3) No classicality collapse. In a topos, internal choice forces Boolean logic (Diaconescu [15, 16]). That argument needs $\vee$, comprehension via the power object, and effective quotients to run — *none* of which are present in the cartesian fragment. Consequently one may add packaging/choice as pure structure without collapsing the object logic to classical. Choosing finite limits is exactly what keeps choice *structural* rather than *logical*. This is the technical heart of the inspectability argument of §1: the choice we add is a nameable operation and it does not silently classicalize everything downstream.

7 The McBride derivative and the proto power object

We now build the object that replaces the power object. The move is to treat every collection *syntactically* — as a term type — and to use the McBride derivative to enumerate its splittings.

The choice of the McBride derivative is not opportunistic. Defining or understanding an object by the totality of its splittings is an intuition that recurs across mathematics, and the McBride derivative is its syntactic, symbolic form. A Dedekind cut defines a real number as a splitting of the rationals into a lower and an upper part; the number *is* the totality of its cuts, an object pinned down entirely by how it comes apart. The covariant derivative in geometry understands a field by splitting its variation along directions, comparing a value against its infinitesimally displaced neighbours; again the object is grasped through its controlled decompositions. In each case an analytic or order-theoretic or differential intuition does the same work: know the whole by the family of its splittings. The McBride derivative [9, 10] performs precisely this move for datatypes — it splits a term into a one-hole context and a focus, and the type of one-hole contexts, the derivative, collects all such splittings — but it performs it on *syntax*,

so the splittings are enumerable, computable, and inspectable. That is exactly what the finite-limits commitment of §6 demands. We adopt the derivative not by loose analogy to cuts and connections but because it is the constructive, symbolic rendering of a move that mathematics has repeatedly found indispensable, and rendering that move syntactically is the whole point.

7.1 Terms, derivatives, and the zipper

Let F be a polynomial (container) endofunctor and let $T = \mu X.FX$ be its initial algebra, the type of terms. Write ∂F for the derivative of F : the functor of one-hole F -contexts [9, 10, 11]. The derivative of the fixpoint is a list of layers up the spine:

$$\partial T \cong (\partial F[T])^* = \text{List}(\partial F[T]),$$

each layer an F -node with one hole and its other children filled by subtrees. Plugging fills the hole:

$$\text{plug} : \partial T \times T \longrightarrow T, \quad \text{plug}(\text{nil}, t) = t, \quad \text{plug}(\ell :: c, t) = \text{plug}(c, \text{fill}(\ell, t)).$$

Definition 7.1 (Splittings). The object of *splittings* of T is

$$\text{Split}(T) = \{ (c, t) \in \partial T \times T \} \cong \sum_{c \in \partial T} T,$$

the pointed terms, with the whole recovered by $\text{plug}(c, t)$. A splitting is a context together with a focus.

Two facts about splittings are load-bearing.

Proposition 7.2 (Positions are ordered by syntax). *The positions of a term — equivalently, the contexts c appearing in its splittings — carry a canonical linear order, the traversal (Dewey) order, definable from the term’s own structure by a fold. Hence ∂T is linearly ordered per term, with no appeal to any order on the atoms.*

Sketch. A depth-first left-to-right traversal assigns to each position a Dewey path, and Dewey paths are linearly ordered lexicographically. The assignment is a catamorphism over T , so it is definable once T is equipped with its recursor (§8); no comparison on atoms is used, because positions are indexed by constructor-argument slots, which are ordered by the arity, not by the atoms sitting in them. $\square$

Proposition 7.3 (The aperture is the modal label). *The contexts that index splittings are the same objects that label rewrites in the behavioral component of §2: both are elements of ∂T . Thus the collection component (via its splittings) and the behavioral component (via its modal labels) are glued at the derivative. In the agency vocabulary, the aperture — the locus at which a computation may be acted upon — is simultaneously the modal label and the raw material of the proto power object.*

Proposition 7.3 is the structural dividend of going through ∂T : components (1) and (3) of §2 are not merely analogous, they share their carrier.

This is the point at which the title earns its keep. The splitting $\text{plug} : \partial T \times T \rightarrow T$, run backwards, is the *axe-stroke*: it cleaves a term into a focus and the context around it, and the aperture ∂T is the opening the stroke leaves. What springs from that opening — fully armed, in the sense that it arrives already carrying the connectives of §2 — is the logic itself. We will call the backward splitting the axe-stroke throughout, and ∂T the aperture it opens; the crucial aspects of logic arise, like Pallas Athena, precisely from separating the focus of attention from its context.

7.2 Collections are term types too

The syntactic stance says: present the collection type itself as a term type, so the same derivative machinery applies to it. The paradigm is the list monad,

$$\text{List } A = \mu X. 1 + A \times X,$$

a genuine μ -type. Its derivative in the element parameter is the pair of a prefix and a suffix,

$$\partial_A(\text{List } A) \cong \text{List } A \times \text{List } A,$$

so a splitting of a list is a triple (prefix, focus, suffix) — exactly a choice of one element to keep in view, with its surroundings. Gathering foci across a family of splittings and re-collecting rebuilds a *subcollection*. This is the concrete content of “no list of all lists”: each individual sublist is free to form, but assembling all of them into one list requires ordering them.

Definition 7.4 (Proto power object). For a term type T , the *proto power object* $\Pi(T)$ is the container of splittings, written $(\partial T, T)$: shapes are contexts $c : \partial T$, and the position at each shape is the focus of type T . Its extension is $\Pi(T)(X) = \sum_{c:\partial T} X$, and $\Pi(T)(T) = \text{Split}(T)$.

$\Pi(T)$ is “proto” because it enumerates one-step focusings; it is not yet closed under gathering foci into a subcollection of the same shape as T . Closing it is packaging.

8 Packaging as a section, and the choice conjecture

8.1 Presenting collections as quotients of the free case

Present each collection monad S syntactically, as a quotient of the free (list) collection:

$$\text{List } A \xrightarrow{q_S} S A,$$

where

$$q_{\text{List}} = \text{id}, \quad q_{\text{Bag}} = (-)/\text{perm}, \quad q_{\text{P}_{\text{fin}}} = (-)/(\text{perm} + \text{idem}).$$

List is the free monoid — the syntactic collection, paying nothing. Bag adds commutativity (permutation). P_{fin} adds commutativity and idempotence. These are precisely the two coordinates of the dial (Principle 3.1) and the two culprits of the no-go theorems (Principle 5.1).

8.2 Two additions before choice

Two things must be added to the bare finite-limits base before packaging can even be discussed, and honesty requires naming them.

Remark 8.1 (The recursor is the first non-Horn addition). Having “the type of terms” means an initial algebra μF , and using it — defining any fold, hence **pack**, hence the Dewey order of Proposition 7.2 — requires its *recursor*. Formation and introduction for a μ -type are algebraic; elimination (induction/recursion) is not cartesian. So the recursor is the first thing added to the bare base, and we add the *weakest* one that still defines **pack**: a non-dependent catamorphism. Everything downstream, including packaging, is a fold and inherits exactly this recursor and no more.

Remark 8.2 (Quotients before sections). For non-free S , the map q_S is a coequalizer, which a bare lex category may lack. Presenting S therefore already commits one to effective quotients (pretopos-flavored structure) *before* asking them to split. Packaging then splits what one was forced to construct.

So the layering is: cartesian base $\rightarrow$ weakest recursor for T (Remark 8.1) $\rightarrow$ effective quotients for non-free $S \rightarrow$ the section below. Choice sits only at the top layer.

8.3 The packaging axiom

Definition 8.3 (Packaging). A *packaging* for S is a signature operation

$$\text{pack}_S : S A \longrightarrow \text{List } A \quad \text{with} \quad q_S \circ \text{pack}_S = \text{id}_{SA},$$

a section of the presenting quotient: a choice, for each S -collection, of a list representative. The axiom asserting the existence of pack_S is PACK_S . In the finite-limits metalanguage PACK_S is not a proposition but the enlargement of the signature by pack_S and the above equation — choice-as-structure, per §6(2).

Definition 8.3 is the syntactic form of the closure operation on the proto power object: to assemble the splittings of $\Pi(T)$ (Definition 7.4) into a S -collection of the same shape is to apply q_S , and to make that assembly canonical — to build $T[(\partial T, T)]$, the collection whose atoms are splittings — is to choose the section pack_S .

8.4 The choice conjecture, parametric in the atom order

Theorem 8.4 (Derivability of packaging, per instance). *Modulo the weakest recursor of Remark 8.1, and for the section taken uniformly (natural in A):*

- (i) $\text{PACK}_{\text{List}}$ is derivable outright; $\text{pack}_{\text{List}} = \text{id}$.
- (ii) PACK_{Bag} is derivable if the atom object A carries a linear order $\leq$ definable in the theory — take $\text{pack}_{\text{Bag}} = \text{sort}_{\leq}$ — and, conversely, a uniform section of q_{Bag} determines a linear order on A (read off from its action on two-element bags). So PACK_{Bag} is inter-derivable with “ A is linearly ordered.”
- (iii) $\text{PACK}_{\text{P}_{\text{fin}}}$ is derivable if A is linearly ordered and equality on A is available to deduplicate — take $\text{pack}_{\text{P}_{\text{fin}}} = \text{dedup} \circ \text{sort}_{\leq}$; conversely a uniform section determines both an order and a canonical representative.

Sketch. (i) is immediate. For (ii), $\text{sort}_{\leq}$ is an insertion sort, a fold over $\text{List } A$ using $\leq$, hence definable from the weakest recursor and $\leq$; it lands in a canonical (sorted) representative of each permutation class, so $q_{\text{Bag}} \circ \text{sort}_{\leq} = \text{id}$. Conversely, given a uniform section s , its value on the bag $\{a, b\}$ is one of the lists $[a, b]$ or $[b, a]$; uniformity (naturality in A) forces this decision to cohere into a total order, and the monoid equations force transitivity. (iii) adds idempotence: a section of the combined quotient must additionally select, from each finite subset, a duplicate-free list, i.e. a decision of equality plus the order of (ii). $\square$

Theorem 8.4 localizes the choice content exactly. Read against the earlier, topos-flavored heuristic in which ordered collections were the expensive case (linearize an unordered carrier =

well-order = full choice), it *inverts* the cost: here **List** is free and the quotients are dear. The two framings concern different maps, and the finite-limits, syntactic stance is what reconciles them:

Principle 8.5 (The McBride move trades and discharges). *Going through the derivative trades the well-ordering flavor of choice for the quotient-section flavor, and then discharges the former for free. The foci of a term-splitting are positions, and positions are pre-ordered by the term's own structure (Proposition 7.2). Syntax hands you the linear order. So over a term rewrite system, packaging is derivable from the positional order, and choice re-enters only through the quotient equations, and only when the atom object A carries no cartesian-definable linear order.*

Conjecture 8.6 (Choice content of packaging). *The choice content of PACK_S equals the assertion that the atom object A admits a linear order. Equivalently: with an order on A supplied, PACK_S is derivable for every finitary S presented as a quotient of **List**; without one, PACK_S is the added choice operation, and its strength is graded by the equations q_S quotients **List** by. In particular the construction is kept parametric in an order on atoms: supplying the parameter makes packaging free; withholding it isolates the entire choice residue at the atom object.*

The parametricity is deliberate and is the hook for later work (§12): in a reflective setting the atoms are quoted processes — names — and an order on names is exactly a place where a knot may be tied. Leaving the order parametric localizes the choice content at the reflection boundary, where we will want to act on it.

9 Grading the conjecture in the Weihrauch lattice

Conjecture 8.6 is not a switch but a dial, and the natural home for its grading is the Weihrauch lattice [17], which measures the uniform strength of a choice principle. We state the intended grading as a program; the finitary cases are within reach of the sketch of Theorem 8.4, the infinitary ones are open.

Conjecture 9.1 (Weihrauch grading). *The Weihrauch degree of PACK_S over an atom object A is monotone in the substructural strength of S — i.e. in the equations q_S adds to **List** — and is bounded as follows:*

- $\text{PACK}_{\text{List}}$: *the identity; the bottom of the lattice.*
- PACK_{Bag} *over finite bags on discrete A : the finite/bounded choice needed to order each finite support uniformly; low in the lattice, below LPO when A is discrete.*
- PACK_{Bag} *over an arbitrary (possibly infinite) atom object: a closed-choice principle C_A on the carrier, climbing toward well-ordering as A grows; the well-ordering flavor of Principle 8.5 re-emerges exactly here, when the syntactic order is unavailable.*
- $\text{PACK}_{\text{P}_{\text{fin}}}$: *the degree of PACK_{Bag} joined with the equality/deduplication decision, adding an LPO-flavored component for the idempotence quotient.*

The value of the Weihrauch reading is that it reframes the no-go theorems of §5 not as yes/no obstructions but as *choice-strength measurements*. Where the classical no-go statement says “no distributive law exists,” the graded statement says “the section that stands in for the missing law has degree d ,” and d is a monotone invariant of how far S sits from the free case. That is the stronger thesis hiding inside the bare conjecture.

10 The three instances, concretely

We collect the worked instances. In each, the theory is the cartesian base extended by the weakest recursor for the relevant μ -type (Remark 8.1); the atom order is the parameter of Conjecture 8.6.

S	equations added to List	pack_S	choice residue
List	none	id	none
Bag	commutativity	$\text{sort}_{\leq}$	order on A
P_{fin}	commutativity, idempotence	$\text{dedup} \circ \text{sort}_{\leq}$	order on A , = on A

Example 10.1 (List: nothing is spent). $\text{List } A = \mu X. 1 + A \times X$; $q_{\text{List}} = \text{id}$; the section is the identity. The proto power object $\Pi(\text{List } A)$ has $\partial(\text{List } A) \cong \text{List } A \times \text{List } A$ (prefix, suffix), and a splitting is (prefix, focus, suffix). Packaging is free because there is nothing to reassemble past the given list order.

Example 10.2 (Bag: order on atoms is spent). $\text{Bag } A = \text{List } A / \text{perm}$. With $\leq$ on A , insertion sort is a fold selecting the sorted representative, and $q_{\text{Bag}} \circ \text{sort}_{\leq} = \text{id}$. Withhold $\leq$ and no cartesian term produces a representative uniformly; pack_{Bag} must be postulated, and by Theorem 8.4(ii) its postulation is inter-derivable with linearly ordering A . This is the cleanest witness for Conjecture 8.6.

Example 10.3 (Finite powerset: order and equality). $P_{\text{fin}} A = \text{List } A / (\text{perm} + \text{idem})$. Packaging needs the sorted representative of the previous example *and* a deduplication, hence decidable equality on A in addition to the order. The two additions are exactly the two quotient equations, confirming Principle 5.1: the idempotence that obstructs P_{fin} 's distributive laws is the same idempotence that must be paid for, as dedup , at packaging time.

11 A payoff: the collection component of Gabbay's coalition logic

We now give the reader a payoff for the formalism. We take a logic that arose independently, for reasons that have nothing to do with realizability — Gabbay's logic of *actionable coalitions*, whose model theory is *semitopology* [18, 19, 20, 21] — and we show that its most distinctive features fall out of the collection-component analysis of §2–§4 as consequences of the shape of a single container. We treat *only* the collection component here; the structural and behavioral components, and the identification of the underlying points with terms of the rho calculus, are the subject of a companion paper.

11.1 Semitopologies, and why they are already a realizability semantics

A *semitopology* [18] is a pair (P, Open) of a set P of points and a set $\text{Open} \subseteq \mathcal{P}(P)$ of open sets with $\emptyset, P \in \text{Open}$ and Open closed under arbitrary unions — but *not* necessarily under intersection. The intended reading is distributed: points are participants, and an open set is an *actionable coalition*, a set of participants with the resources to take a collaborative action without the permission of anyone outside it. A union of actionable coalitions is actionable; an intersection need not be (two coalitions each strong enough to act may overlap in a single participant, who is not). This one dropped axiom is the whole of the difference from topology.

What matters for us is that Gabbay motivates the definition in exactly the terms of §1. Following the observational tradition in which opens model affirmations, or semidecidable properties [22], he reads an open set as an *actionable affirmation*: an outcome that can be agreed and enacted within the coalition. Topology, on this reading, is about what can be positively *said*; semitopology is about what can be positively *done*. An open is thus already a collection of computational witnesses — participants who can jointly realize an outcome — which is to say that semitopology is a realizability semantics whose container is the opens of a semitopology. This is precisely the setting our collection-component analysis was built to read.

11.2 The connectives fall out of the container

Fix a semitopology (P, Open) and take the collection container to be $S = \text{Open}(P)$, ordered by $\subseteq$, with a formula denoting the actionable coalition of participant-witnesses that realize it. Running the collection component of §2 — connectives are exactly the operations under which the container is closed — gives the following, with nothing added by hand.

Proposition 11.1 (Connectives of the semitopological collection component). *Let $S = \text{Open}(P)$ for a semitopology (P, Open) . Then:*

- (i) *S is closed under arbitrary unions, so disjunction $\bigvee = \bigcup$ is a total, arbitrary (geometric) collection operation: a genuine collection connective.*
- (ii) *S is in general not closed under binary intersection, so conjunction is not interpretable as a collection operation on S . The largest coalition below two opens, $\text{int}(A \cap B) = \bigcup\{U \in S : U \subseteq A \cap B\}$, exists by union-closure but is a derived operation — an interior, i.e. a closure — not the container's intersection.*
- (iii) *The primitive relational residue of conjunction is therefore the compatibility relation*

$$A * B \iff A \cap B \neq \emptyset,$$

which records co-realizability — the existence of a witness realizing both — without producing a witness-collection.

Proof. (i) is closure under arbitrary unions, an axiom of semitopology. (ii): if Open were closed under $\cap$ it would be a topology; a semitopology is exactly one that need not be, and $\text{int}(A \cap B)$ is the greatest open below $A \cap B$ by union-closure (the union of all opens contained in $A \cap B$). (iii) is immediate: $*$ is definable but is a relation, not an operation into S . $\square$

The content of Proposition 11.1 is the asymmetry it exposes. On the substructural dial of Principle 3.1, the aggregation of this container is union, which is commutative and idempotent, so on the *join* the logic looks fully structural. But the dial's two coordinates do not see the phenomenon that actually distinguishes this setting, which is that *disjunction is container-primitive while conjunction is not*. This is a finer axis than the two Booleans of §3:

Principle 11.2 (Which lattice operations are container-primitive). *Beyond which structural rules hold (Principle 3.1), a container is graded by which lattice operations it makes primitive. A topology makes both arbitrary $\bigvee$ and finite $\wedge$ primitive: its opens form a frame, and the collection component yields geometric/intuitionistic logic. A semitopology makes only $\bigvee$ primitive: its opens form a semiframe, and the collection component yields a weaker logic of actionable coalitions in which conjunction survives only relationally, as $*$, or derivatively, as $\text{int}(- \cap -)$.*

11.3 The framework predicts the semiframe

Here is the payoff proper. The algebraic dual of a topology is a frame: a complete lattice in which finite meets distribute over arbitrary joins. The algebraic dual of a semitopology cannot be a frame, because Proposition 11.1(ii) has removed the meet. What must replace it? Exactly the relational residue of Proposition 11.1(iii).

Observation 11.3 (The compatibility relation is forced). *The collection-component analysis predicts, from the closure properties of the container alone, that the algebraic dual of a semitopology is a complete join-semilattice equipped not with a meet operation but with a symmetric, join-distributive compatibility relation $*$ abstracting nonempty intersection. This is exactly Gabbay’s semiframe [20]: a complete join-semilattice $(X, \leq)$ with a relation $*$ $\subseteq X \times X$ that is symmetric, properly reflexive ($x * x \Leftrightarrow x \neq \perp$), and completely distributive over joins ($x * \bigvee X' \Leftrightarrow \exists x' \in X'. x * x'$). Gabbay arrived at $*$ independently, as the structure needed to make the semitopology–semiframe duality go through and to express the anti-separation (intertwinedness) conditions that govern consensus; the collection component recovers the same relation as the forced residue of a meet-free container. Two routes, one relation.*

Observation 11.3 is the concrete cash value of the abstract disconnect of §4. There we said that categorical logic packages the collection component through a device — the subobject classifier, equivalently the frame of subobjects — that bakes in set-shaped, meet-closed collection, and that the packaging fails once the container is not meet-closed. Semitopologies are exactly such a container, and the failure is not pathology but the whole point: in a decentralised system the intersection of two actionable coalitions is not actionable, so the logic of the system *must* be the meet-free one. The seam that §4 exposed abstractly is, here, the seam along which distributed consensus is actually cut.

11.4 What the residual structure means under realizability

Reading Gabbay’s constructions back through the realizability lens gives them a computational gloss worth recording, since it is what the companion paper will build on.

The compatibility relation $A * B$ says that $\llbracket \varphi \rrbracket$ and $\llbracket \psi \rrbracket$ share a witness: φ and ψ are co-realizable, even when $\varphi \wedge \psi$ has no direct reading as a coalition. This is strictly weaker than demanding that the meet be a nonempty collection, and it is the right primitive: in a distributed setting one wants to know whether two actions *can* be taken by an overlapping party, not to name the coalition of all parties who take both.

A *topen* — a transitive open set, the pivotal notion in Gabbay’s consensus theory — is an open T such that any two opens meeting T meet each other. Under realizability this is a witness-collection with internal coherence: on a topen, continuous value assignments are forced to agree [18], so a topen is a collection of witnesses *guaranteed to decide alike*. A *regular* point is a participant lying in such a coherent community. In the companion paper, where points are rho-calculus processes, a topen becomes a set of processes guaranteed to reach consensus, and the collection component developed here becomes the propositional logic of actionable coalitions of processes.

11.5 Why the finite-limits commitment is repaid here

There is a final dividend, and it ties directly to §6. Moving from a topology to a semitopology moves conjunction from an *operation* $\wedge : X \times X \rightarrow X$ subject to the frame distributive law,

to a *relation* $* \subseteq X \times X$ subject to three Horn-shaped axioms. A relation is, in a finite-limits metalanguage, the most native thing there is: a subobject $* \rightarrowtail X \times X$ carved out by equations, with the join-distributivity axiom a geometric implication of exactly the kind cartesian logic is built to state. So dropping intersection-closure does not merely model distributed systems; it lands the collection component in a more purely relational, more cartesian presentation than the topological case. The semitopological setting is *more* syntactic than the topological one, not less — precisely the direction the finite-limits commitment of §1 and §6 was pushing us.

Remark 11.4 (Compatibility is a proto-conjunction). The parallel with the proto power object (§7) is worth one line. The compatibility relation $*$ is to conjunction what the object of splittings is to the power object: the un-packaged, relational form that records possibility without producing the packaged object. Recovering an actual conjunction from $*$ means applying the interior — a closure, a global operation — just as recovering a subcollection from splittings meant applying **pack**. Conjunction-as-relation versus conjunction-as-operation is one more instance of proto versus packaged.

12 Discussion

No classicality collapse. The absence of Diaconescu’s argument in the cartesian fragment (§6(3)) is what makes this entire program viable. We add a choice operation — **pack_S** — and the object logic does not thereby go Boolean, because the collapse needs $\vee$, comprehension, and effective-quotient machinery that the metalanguage does not provide. Choice remains structural. A logic assembled this way can have any setting of the substructural dial (Principle 3.1) and still admit packaging, without being forced to classicalize. This is the payoff we most want the AI community to register: inspectable choice that does not contaminate the logic it serves.

Quotient versus choice as dual dodges. §5 noted two standard escapes from the no-go theorems — quotient (adjust S until the equations agree) and restrict (shrink until they cannot clash) — and offered a third, adding a global section. We conjecture these are dual: the quotient-dodge builds a coequalizer, the section-dodge splits it; the restrict-dodge builds a well-founded (finite, ordered) carrier, the section-dodge well-orders it. In slogan form: *coequalizer versus Zorn*. Making this duality precise — exhibiting the section-dodge as the formal dual of the quotient-dodge in the appropriate fibration — is the natural next technical step, and it would situate the construction cleanly against the existing distributive-law literature.

Graphs are the genuine frontier. Everything here rests on T being a polynomial/container functor, so that ∂T exists and Definition 7.4 is available. Lists and trees are fine. Graphs are not free/regular in the naive sense, the McBride derivative does not directly apply, and the proto power object is not defined. We bracket the graph case explicitly rather than let it ride along with lists and trees; adapting the derivative to non-regular carriers (or replacing it by a suitable comonadic/traced structure) is the honest open problem for extending the collection dial past the container-shaped settings.

The atom-order parameter and knot-tying. Conjecture 8.6 keeps the order on atoms parametric on purpose. In the reflective (ρ) setting the atoms are quoted processes — names — and the boundary between process and name is exactly the reflection boundary. An order

on names is therefore a structure imposed *at* that boundary, and it is precisely the datum a knot-tying construction will manipulate. By isolating the entire choice residue of packaging in the atom order, the present construction leaves the knot exactly one place to be tied: supply the order (and packaging trivializes; the knot is untied) or withhold it (and the choice residue is the knot). This is the sense in which the finite-limits, derivative-based account is not merely a reconciliation of the collection component with categorical logic, but a setup for the reflective constructions to come.

Summary. The three-component decomposition makes the seams of a logic visible under realizability; the collection component is a substructural dial governed by commutativity and idempotence; those same two equations obstruct the distributive laws and set the cost of packaging; the finite-limits metalanguage removes the subobject classifier (forcing the syntactic route) and forbids stating choice as a proposition (forcing it to be a named section), while blocking the classicality collapse that would otherwise make this untenable; the McBride derivative supplies the proto power object and, through the positional order on splittings, discharges the well-ordering flavor of choice; and what remains — the choice content of packaging — is exactly the order on atoms, kept parametric for the knot to act on. That is the construction this programme set out to formalize, and the shape of the program it opens.

13 Coda: collections have computational content

One point remains, and it is the one we most want to leave with the reader, because it dissolves a confusion that has dogged program extraction for decades. A collection is not a neutral container. A collection type carries computational content in and of itself, and so the realizability content $\llbracket \varphi \rrbracket : S(T)$ has its computation *parsed* between the collection S and the explicit computational ontology T . Neither component alone fixes the computational model, and until the parse is acknowledged, extraction and its dual are ill-posed.

13.1 The collection’s equations are a computational ontology

Read the quotient tower of §8 once more, now for its computational content rather than its choice content:

- **List** (ordered, multiplicities retained) is *sequential* computation: traces, iteration, the fold as the machine.
- **Bag** (commutativity added) is *concurrent, resource-sensitive* computation: multiset rewriting, the chemical abstract machine [24], the commutative parallel composition of a process calculus.
- **P_{fin}** (idempotence added) is *nondeterministic, relational* computation: choice with neither order nor multiplicity.
- **D** (convexity in place of idempotence) is *probabilistic* computation.

Each quotient in the tower forgets a layer of computational content: permutation forgets sequencing, idempotence forgets multiplicity. The substructural dial of §3 is therefore, read from this side, a dial on *how much computation the collection itself carries*, and the packaging quotients of §8 are exactly the maps that discard it.

Principle 13.1 (The dial is a computational ontology). *The algebraic theory of S is not merely a set of structural rules; it is a computational ontology. The tower $\text{List} \rightarrow \text{Bag} \rightarrow \text{P}_{\text{fin}}$ is a spectrum of progressive forgetting — sequential, then concurrent, then nondeterministic — and where a collection sits on it fixes the computational model it can natively express.*

13.2 Computation is conserved and parsed across (S, T)

A domain-specific computational model must be recovered from the whole of $\llbracket \varphi \rrbracket : S(T)$, and its content is apportioned between the two components. Coarsen S — move up the tower, forget — and the forgotten content must reappear in T if the same model is to be recovered; enrich S and T may be left bare. The extremes and the middle are all inhabited:

- *All in S :* T degenerate, the model is the collection’s native one — pure nondeterminism is P_{fin} over trivial terms.
- *All in T :* S a bare set, every domain-specific feature carried explicitly by the rewrite system — the default posture of classical realizability, and the source of the confusion below.
- *The middle*, where real systems live: a process calculus places its concurrency in a bag-shaped collection (parallel composition is commutative) and its reflection and higher-order structure in T . This is the parse the companion paper adopts, with T the rho calculus [3].

Principle 13.2 (Conservation and parsing). *The computational content of $\llbracket \varphi \rrbracket : S(T)$ is conserved and parsed across the pair (S, T) . Content forgotten by S must be supplied by T , and content carried by S need not be duplicated in T . No single component fixes the model; the pair does.*

13.3 Why extraction was confused: the Olympian’s headaches

Now the confusion — the headaches that, in the myth, precede the stroke. The proofs-as-programs tradition typically fixes S at the set/powerset end — propositions as sets of realizers — and then expects the extracted program to inhabit an arbitrary, often sequential and Turing-complete, computational model. But a set or powerset collection has *already* quotiented out order and multiplicity (Principle 13.1). Extraction through it cannot recover sequencing or resource-sensitivity; it recovers only the nondeterministic-relational shadow that survives the quotient.

Observation 13.3 (Collections do not recover arbitrary models). *A collection, without additional information, cannot recover an arbitrary computational model. It recovers precisely the models native to its ontology. To extract a domain-specific program one must either move S down the tower until it retains the needed content, or carry that content explicitly in T ; expecting a fixed collection to deliver an arbitrary model is the category error at the root of the extraction confusion.*

13.4 Extraction and classification are dual

The parse governs not one procedure but two, dual to one another. Write the realizability relation as $\Vdash \subseteq \text{Comp} \times \text{Form}$. Its two transposes are:

$$\text{extraction: } \varphi \longmapsto \llbracket \varphi \rrbracket \quad (\text{synthesis; formula to its witnesses}),$$

classification: $e \longmapsto \{\varphi : e \Vdash \varphi\}$ (analysis; computation to the formulae it realizes),

adjoint in the manner of a Galois connection. The parse (S, T) sets the resolution of both at once: coarsening S lowers the resolution of extracted programs and, in the same motion, coarsens classification, so that formulae separate computations less finely. The granularity is shared because it is one relation read in two directions. Consequently, fixing the parse makes both procedures effective: one knows which model the extracted program inhabits (the native model of S , refined by T), and at what granularity witnesses are classified. That is what it means to understand extraction and witness classification effectively.

Observation 13.4 (Semitopology, once more). *The payoff of §11 is an instance. The open-set collection has semidecidability — affirmability, actionability — as its native ontology, so extraction from an open is a semidecision procedure and classification is the assignment to a participant of the affirmations it realizes, i.e. the coalitions it joins. A topen is then a classification guaranteed to be consensus-coherent: a family of witnesses that must be classified alike. The semitopology is simply the choice to parse observational content into S .*

13.5 The last seam

There is a reason this could only be said cleanly from a finite-limits metalanguage. Keeping T explicit and syntactic (§6) is what makes the parse legible; hide T inside an ambient model — a topos, a fixed Ω , an assumed Turing-complete substrate — and the division of computational content between collection and terms becomes invisible, which is exactly how the extraction confusion took hold. The three components of §2 were the seams of a logic. The parse is the seam within computation itself: between what the collection already computes and what the terms make explicit. Making that seam visible was the whole point of the programme, and it is what turns program extraction and witness classification from standing sources of confusion into effective, dual procedures. The headaches lift when the stroke falls: it is the splitting of focus from context that lets the logic spring out whole.

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